Comparing High-Dimensional Forecasting Methods: The Case of Oil Price Prediction

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1 Introduction and Motivation

The objective of our study is to determine whether different forecasting methods can predict monthly changes in WTI crude oil prices when using a broad range of economic and financial indicators, and to assess their predictive effectiveness. In addition, we want to understand how dimension-reduction techniques and regularization contribute to forecast accuracy when the data are high-dimensional and the underlying economic environment is relatively complex.

We integrated multiple indicators reflecting the state of the U.S. economy and global energy market dynamics, including data from FRED and the World Energy Overview. This issue is critical because oil price volatility impacts inflation levels, production costs, and financial markets, holding significant implications for policymakers, investors, and energy-intensive enterprises. Furthermore, existing research indicates that sharp oil price fluctuations often transmit significantly to inflation, a phenomenon particularly pronounced in developed economies and energy-importing nations (Ha et al., 2023).

One challenge is the large number of predictors relative to the sample size, which is the so-called "curse of dimensionality". When this happens, classical regression techniques such as OLS often overfit and produce unstable estimates, and it requires methods that are better suited for handling many covariates at once. So we evaluate different forecasting models using a rolling window of 100 periods, with the random walk serving as the baseline model for comparison. To be clear, we group the models as follows:

1. **Unsupervised factor models:** PCR, DFM
2. **Supervised factor models:** 3PRF, PCLAR, PCLAS
3. **Regularization methods:** LASSO, Ridge, Elastic Net

We include a simple random walk model as benchmark. Upon this, we consider factor-based methods such as Principal Component Regression (PCR) and Dynamic Factor Models (DFM), which summarize a large group of predictors into a smaller set of latent factors, time-series dynamics are also taken into account in the case of DFM (Bai and Ng, 2002; Babura et al., 2015; Doz et al., 2012). We also examine supervised factor methods, including the Three-Pass Regression Filter (3PRF) and two Principal Component LASSO variants (PCLAR and PCLAS), which are constructed to focus specifically on the part of the factor space that is relevant for predicting oil prices (Kelly and Pruitt, 2015; Bai and Ng, 2008). Last, we make use of regularization methods, including LASSO, Ridge, and Elastic Net, to help stabilize estimation by shrinking or selecting coefficients when the number of predictors is large.

The paper is structured as follows. Section 2 describes the source and process of data. Section 3 introduces methodology we apply. Section 4 reports empirical results for our selected methods' forecasts. Section 5 discusses and concludes.

2 Data

2.1 Data Source

Our analysis relies on a comprehensive dataset that combines U.S. macroeconomic indicators with global energy market variables. All series are monthly, covering the period from January 1995 to December 2019 after processing. This period was chosen to balance data availability and economic representativeness. It contains key historical events such as the late-1990s oil market stabilization, the mid-2000s commodity boom, the 2008–2009 financial crisis, and

the 2014–2016 oil price collapse. We exclude the highly volatile 2020–2021 COVID-19 period to avoid distortions in typical oil price dynamics. Meanwhile, data from both FRED-MD and international energy sources are consistently available from the mid-1990s, making 1995 a natural starting point for constructing a unified panel.

2.1.1 Core and Target Variables

Our core predictor set consists of over 100 U.S. macroeconomic and financial indicators drawn from the FRED-MD dataset, covering industrial production, employment, housing activity, consumer prices, interest rates, and various financial market measures. Together, they provide a relatively comprehensive picture of the U.S. economy that is relevant for forecasting oil prices.

The target variable is the West Texas Intermediate (WTI) crude oil price (`POILWTIUSDM`), measured in U.S. dollars per barrel.

2.1.2 External Energy Market Variables

To capture broader energy market conditions, we supplement the core predictors with selected variables from the World Energy Overview(WEO) and FRED datasets. We kept initially only two variables[1] from WEO due to collinearity between variables among the dataset:

- **Brent Crude Oil Price (POILBREUSDM)** – a global benchmark.
- **U.S. Henry Hub Natural Gas Price (PNGASUSUSDM)** – representing U.S. natural gas markets.
- **Diesel Producer Price Index (WPS057303)** – tracking domestic diesel price changes.
- **Industrial Production: Mining – Crude Oil (IPG21112S)** – capturing U.S. crude oil extraction activity.
- **Primary Energy Net Imports and Primary Energy Stock Change and Other** – reflecting U.S. energy supply-demand balance and inventory adjustments.

Overall, this high-dimensional panel integrates macroeconomic, financial, and energy indicators, providing a solid foundation for evaluating different forecasting methods. In the following sections, we will examine how these variables contribute to predicting WTI crude oil prices.

2.2 Data Transformation and Diagnostic Checks

To ensure the validity of the forecasting exercise, we applied a series of steps to process the dataset before model estimation. First, the stationarity of all variables was assessed using the Augmented Dickey–Fuller (ADF) test, where the null hypothesis corresponds to the presence of a unit root. Each of the 134 (including Y) numeric time series was tested individually, and a variable was considered stationary if its p-value was below 0.05. At their levels, 119 variables were identified as non-stationary ($I(1)$), while the remaining series were already stationary. To induce stationarity, we computed first differences for all non-stationary variables, after which the ADF test was re-applied. There were fourteen variables that remained non-stationary, and thus we chose to difference a second time. Following the second differencing, all variables achieved stationarity and were ready for modeling.

Following the stationarity adjustments, correlation diagnostics were conducted to prevent target leakage. Pairwise correlations were computed between each predictor X_j and the dependent variable Y_t (the WTI crude oil price). Using a conservative correlation threshold of 0.9, two predictors were found to be excessively correlated with the target: **OILPRICEx** (Crude Oil, spliced WTI and Cushing) with a correlation of 0.999, and **POILBREUSDM** (Global price of Brent Crude) with a correlation of 0.936. Since both variables essentially measure the same thing as the target series, including them would create target leakage and boost forecasting performance. These variables were therefore excluded from the modeling dataset prior to estimation. Consequently, there are 131 variables in our predictor matrix X_t.

[1] We ultimately retain only two variables: **Primary Energy Net Imports** and **Primary Energy Stock Change and Other**. These indicators reflect the external balance and inventory adjustments of the U.S. energy system, providing direct information on supply and demand dynamics while avoiding redundancy with other production or consumption measures.

As a final step, we also examined the potential redundancy among predictors. For this purpose, we applied a multicollinearity check based on QR decomposition [2], which showed that four variables were almost linear combinations of the others. To understand whether this issue had practical implications, we estimated several preliminary specifications both including and excluding these variables. The main outcomes, such as coefficient patterns, overall fit, and general stability, remained almost the same across all versions of the model.[3] Removing variables did not offer a clear advantage, because the forecasting methods used in our analysis, like regularization procedures and factor models, already account for multicollinearity by shrinking coefficients or extracting orthogonal components. Thus, we decided to retain the full set of predictors to preserve the full information set.

2.3 Descriptive Statistics

Table 1 summarizes the descriptive statistics for the WTI crude oil price series after first differencing (our target variable Y) over the baseline sample from January 1995 to December 2019. The mean monthly change is 0.1422, while the median is 0.6251, indicating that typical monthly changes tend to be slightly larger than the average change. The standard deviation of 4.8832 indicates substantial volatility, consistent with the presence of large shocks during the sample period.

The minimum and maximum monthly changes are -28.1597 and 13.5217, respectively, highlighting extreme movements such as those observed during the 2008 financial crisis. The skewness of -1.3215 suggests a left-skewed distribution, meaning large negative shocks occur more frequently than large positive ones. Furthermore, the kurtosis of 8.3133 indicates heavy tails, implying that extreme events are far more likely than under a normal distribution. These characteristics confirm that oil price dynamics are highly volatile and non-normal, making forecasting particularly challenging and motivating the use of robust high-dimensional methods.

Figure 1 shows the monthly changes in WTI crude oil prices over 1995 to 2019. The series exhibits pronounced fluctuations throughout the sample, with sharp declines during the 2008 financial crisis and multiple large rebounds in the subsequent years. This high volatility and structural variation underscore the complexity of the forecasting task.

Table 1: Descriptive Statistics for WTI Oil Price after differencing (Y)

Mean	Median	SD	Min	Max	Skewness	Kurtosis
0.1422	0.6251	4.8832	-28.1597	13.5217	-1.3215	8.3133

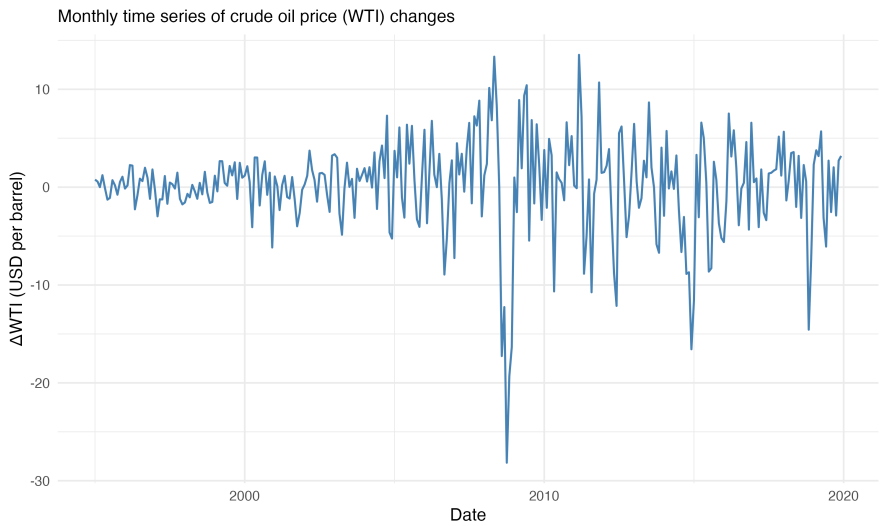

Figure 1: Monthly changes in WTI crude oil price (1995–2019)

[2] QR decomposition is a matrix factorization method used here to diagnose multicollinearity. When the predictor matrix X is decomposed into $X = QR$ (where Q is an orthogonal matrix and R is an upper triangular matrix), strong multicollinearity is indicated by small diagonal elements in the R matrix, signifying near-linear dependence among the predictors.

[3] The project folder includes an R script where the forecasting procedure is re-estimated after removing redundant predictors. The results are almost unchanged.

3 Methodology

The main aim of this study is to tackle the challenges of forecasting WTI crude oil prices using high-dimensional data. In our dataset, the number of predictors (N) exceeds the number of observations (T) in each 100-period rolling window. This situation creates two main statistical issues.

A large number of predictors can cause the curse of dimensionality. When N is high, traditional methods like ordinary least squares (OLS) suffer from high variance. This lowers out-of-sample forecasting performance and can make model estimation difficult. At the same time, predictors are often highly correlated, leading to strong multicollinearity. Multicollinearity makes coefficient estimates unstable and reduces both interpretability and reliability.

To handle these challenges and maintain robust forecasting, we use a fixed-length 100-period rolling window as our main evaluation framework. Within this setup, we apply three groups of forecasting methods on top of a baseline model: (1) unsupervised models, (2) supervised models, and (3) regularization techniques. Because these methods vary in complexity and require careful tuning of key parameters, we adopt two complementary criteria for model selection: the Bai and Ng IC_2 for factor models and the Bayesian Information Criterion (BIC) for regularization models.

3.1 Forecasting Framework: Rolling Window

This study evaluates the out-of-sample performance of the forecasting models using a rolling window framework. This approach mimics real-time prediction, where parameter estimates are updated as new data arrive (Tashman, 2000). It also lets the model adjust to changes in the data rather than relying on one long fixed sample.

Let T be the total number of observations and set the window length to $R = 100$. At each step the model is estimated on the most recent R observations, $(Y_t, \mathbf{X}_t)_{t=i}^{i+R-1}$, and a one-step-ahead prediction $\hat{Y}_{i+R}$ is produced. The window then moves forward by one period and the procedure is repeated until the end of the sample. In total this generates $P = T - R$ forecasts. Forecast errors are defined as $e_{i+R} = Y_{i+R} - \hat{Y}_{i+R}$, and these errors form the basis for comparing models.

The decision to use a 100-month window reflects both statistical needs and practical considerations. With 131 predictors, the model requires a reasonably long estimation sample for extracting factors, yet it must also remain sensitive to structural changes in the oil market. A window of 100 months, roughly eight years, captures a full business-cycle horizon but is still short enough to react to shifts in market conditions. The rolling window is preferred over expanding or fixed windows because it allows parameter estimates to adjust to new information while keeping the computational burden stable. An expanding window may give too much weight to old data. A fixed window, on the other hand, may drop useful information too quickly. The rolling approach strikes a middle ground and helps reduce the impact of structural breaks in the data-generating process.

The analysis focuses on one-step-ahead (one-month-ahead) forecasts. This horizon is directly relevant for decision-making and provides a consistent benchmark across models. Longer horizons would accumulate errors and are especially problematic when the underlying series is highly persistent (with $\rho \approx 0.95$).

Forecast accuracy is evaluated using standard loss functions, including the mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE):

$$\mathrm{MSE} = \frac{1}{P}\sum_{t=R+1}^{T} e_t^2, \quad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \quad \mathrm{MAE} = \frac{1}{P}\sum_{t=R+1}^{T} |e_t|. \tag{1}$$

In addition, out-of-sample explanatory power is measured by

$$R^2_{\mathrm{OOS}} = 1 - \frac{\sum_{t=R+1}^{T} e_t^2}{\sum_{t=R+1}^{T} (Y_t - \bar{Y})^2}, \tag{2}$$

where $\bar{Y}$ denotes the mean of the out-of-sample realizations. A positive R^2_{OOS} indicates improvement over the historical mean benchmark. To facilitate comparison with the random walk model, we also report the relative RMSE:

$$\text{Relative RMSE} = \frac{\mathrm{RMSEmodel}}{\mathrm{RMSERW}}, \tag{3}$$

where values below one signify gains relative to the naïve forecast.

All models are re-estimated within each rolling window to avoid look-ahead bias. Predictors are standardized using the means and standard deviations from the corresponding estimation window, while the dependent variable is left unstandardized for ease of interpretation. For factor-based models, including PCR, DFM, 3PRF, PCLAR, and PCLAS, the number of factors r is selected using the Bai and Ng (2002) IC_2 criterion. For regularization methodsRidge, LASSO, and Elastic Netthe penalty parameter λ is chosen by minimizing the Bayesian Information Criterion (BIC) within each window. Models that incorporate lagged dependent variables adjust the effective information set accordingly: 3PRF uses six lags of Y, whereas DFM, PCLAR, and PCLAS include a single lag [4].

3.2 Model Selection Criteria

3.2.1 Bai & Ng IC_2 for Factor-Based Models

For models that rely on latent factor extraction, such as Principal Component Regression (PCR), Dynamic Factor Model (DFM), Three-Pass Regression Filter (3PRF), Partial Component Least Angle Regression (PCLAR), and Partial Component Least Absolute Shrinkage (PCLAS), the number of common factors is determined using the IC_2 criterion proposed by Bai and Ng (2002).

Given the predictor matrix X_t of dimension $N \times T$, let V_k denote the average residual variance when X_t is approximated by k common factors. The IC_2 criterion is defined as

$$\mathrm{IC}_2(k) = \ln(V_k) + k \cdot g(N, T),$$

where the penalty term $g(N, T)$ is given by

$$g(N, T) = \frac{NT}{N + T} \ln(\min(N, T)).$$

This criterion provides a balance between model fit and simplicity. The first term rewards better fit, while the second term penalizes excessive model complexity. For each 100-period rolling window, we calculate $\mathrm{IC}_2(k)$ for $k = 1, \ldots, k_{\max}$, where $k_{\max}$ is set to $\min(10, \lfloor T/5 \rfloor)$ to avoid overfitting. The optimal number of factors, denoted as $\hat{k}$, is selected as the value of k that minimizes $\mathrm{IC}_2(k)$. This adaptive procedure allows the number of factors to vary over time, enabling the model to respond to structural changes in the data and maintain consistent forecasting performance.

3.2.2 Bayesian Information Criterion (BIC) for Regularization-Based Models

For the regularization methods, which include Ridge, LASSO, and the Elastic Net, the key step is choosing the regularization parameter λ. This parameter controls how strongly the coefficients are penalized. In practice, the model needs a balance between fit and simplicity, so λ is picked by minimizing the Bayesian Information Criterion (BIC) (Schwarz, 1978) within each 100-period rolling window.

$$\mathrm{BIC}_\lambda = T \cdot \log(\mathrm{MSE}_\lambda) + d_\lambda \cdot \log(T), \tag{4}$$

where T is the number of observations in the training sample and MSE_λ is the mean squared error for that specific λ. The term d_λ reflects the effective degrees of freedom of the penalized model. Using BIC keeps the procedure computationally light, which matters in large rolling-window exercises [5].

The degrees of freedom, d_λ, depend on the type of penalty. For Ridge, it is given by the trace of $\mathbf{X}(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}'$. For LASSO, d_λ simply equals the number of non-zero coefficients, highlighting its built-in variable selection. The Elastic Net sits in between, as its effective degrees of freedom combine features of both L_1 and L_2 penalties.

Once the BIC is evaluated for all candidate values, the λ that minimizes BIC_λ is selected, and the corresponding parameter estimates are used to produce the one-step-ahead forecast.

[4] The selection of six lags for 3PRF follows the recommendation by Kelly and Pruitt (2015), which suggests a number of lags sufficient to capture the predictors' sensitivity to past movements of the target variable. The single lag for DFM, PCLAR, and PCLAS is chosen to capture the most direct, short-term time-series persistence.

[5] The choice of BIC over more computationally intensive methods like k-fold cross-validation is motivated by the need for computational efficiency in a large-scale, repeated rolling-window forecasting exercise. BIC is known to be effective for large-sample model selection and avoids the heavier resampling required by cross-validation.

3.3 Baseline Model: Random Walk Model

In empirical studies of time series forecasting, the Random Walk (RW) model is commonly used as the minimum benchmark to assess whether any complex model possesses economic value and predictive power. If a sophisticated model designed to exploit high-dimensional macroeconomic factor information fails to consistently outperform the RW model, it suggests that the information used (i.e., macroeconomic factors) is ineffective for the forecasting task, or that the model structure is flawed. This study adopts the RW model as the baseline for evaluating the predictive performance of all macro factor models and dimensionality reduction methods. The Random Walk (RW) benchmark used in this study corresponds to the standard assumption that the price level P_t follows a random walk:

$$P_{t+1} = P_t + \varepsilon_{t+1}, \tag{5}$$

where ε_{t+1} is an unpredictable innovation. Since the dependent variable in our forecasting models is the first difference of the price series, $Y_t = \Delta P_t$, the RW benchmark for Y_t reduces to a simple last-observation forecast:

$$\widehat{Y}_{t+1|t} = Y_t. \tag{6}$$

This formulation matches the empirical implementation, where the next-month forecast is set equal to the most recently observed monthly change in WTI prices. The RW model therefore provides a rigorous naive benchmark: factor-based and penalized regression models must achieve lower forecast errors than this benchmark to demonstrate incremental predictive value.

3.4 Unsupervised Factor Model

3.4.1 Principal Component Regression (PCR)

Principal Component Regression (PCR) is a dimension-reduction forecasting approach based on principal component analysis (PCA). The method separates the extraction of common factors from the predictive regression. This structure helps reduce multicollinearity and limits overfitting in high-dimensional environments. PCR is implemented in a rolling-window setting, and the entire procedure is re-estimated in each window of 100 observations.

Within every window, PCA is applied to the predictor matrix $\mathbf{X}_t$. The number of retained factors k is chosen using the IC_2 criterion of Bai and Ng (2002). The selected principal components

$$\mathbf{F}_t = (F_{1,t}, F_{2,t}, \ldots, F_{k,t})',$$

represent the estimated common factors.

Once the factors are obtained, the forecasting regression is estimated by regressing Y_t on $\mathbf{F}_t$. The one-step-ahead forecast $\widehat{Y}_{t+1|t}$ is constructed by projecting the next-period predictor vector $\mathbf{X}_{t+1}$ onto the factor loadings, which yields the predicted factor scores $\hat{\mathbf{F}}_{t+1|t}$:

$$\widehat{Y}_{t+1|t} = \hat{\alpha} + \hat{\boldsymbol{\beta}}'\hat{\mathbf{F}}_{t+1|t}.$$

In this setting, α is the intercept and $\boldsymbol{\beta}$ collects the coefficients that map each factor to the target variable. At each forecasting step, both the factor structure and the regression coefficients are re-estimated, and the prediction relies on factor scores computed from the projected new observations.

3.4.2 Dynamic Factor Model (DFM)

The Dynamic Factor Model (DFM) builds on the PCR framework. In addition to the contemporaneous macroeconomic factors, it allows the target variable Y_t to depend on its own lag. This specification, often referred to as a Factor-Augmented Autoregressive (FAAR) model, evaluates whether including time-series persistence improves forecast accuracy.

As in PCR, the factor extraction is performed separately in each rolling window and the number of factors k is selected using the IC_2 criterion. After extracting $\mathbf{F}_t$, the forecasting equation adds the first lag of the target variable:

$$Y_t = \alpha + \rho Y_{t-1} + \boldsymbol{\delta}'\mathbf{F}_t + u_t.$$

Here, α denotes the intercept. The parameter ρ measures the autoregressive effect of the previous period's oil price change. The vector $\boldsymbol{\delta}$ captures the contribution of the macroeconomic factors.

The one-step-ahead forecast relies on factor scores obtained by projecting $\mathbf{X}_{t+1}$ onto the factor loadings from the current window. Denoting these scores by $\hat{\mathbf{F}}_{t+1|t}$, the forecast is

$$\widehat{Y}_{t+1|t} = \hat{\alpha} + \hat{\rho} Y_t + \hat{\boldsymbol{\delta}}' \hat{\mathbf{F}}_{t+1|t}.$$

This specification allows the model to exploit both macroeconomic fundamentals and the intrinsic persistence of the price series. Comparing the performance of DFM and PCR provides evidence on whether, in a high-dimensional forecasting environment, predictive accuracy is driven mainly by common factors (PCR) or by models that also incorporate time-series dependence in oil price changes (DFM).

3.5 Supervised Factor Model

Supervised Dimension Reduction (SDR) extends traditional factor-based approaches such as PCR and DFM by embedding the forecasting objective directly into the factor extraction process. Unlike unsupervised methods that compress the predictor matrix $\mathbf{X}$ by maximizing variance, SDR uses the target variable Y to identify the most predictive directions. In this way, the method extracts components that are most informative for forecasting Y, making it particularly suitable when the predictors most relevant to Y are not the principal drivers of variation in $\mathbf{X}$. Two representative supervised approaches are considered: 3PRF and PCLAR/PCLAS.

3.5.1 Three-Pass Regression Filter (3PRF)

The Three-Pass Regression Filter (3PRF) proposed by Kelly and Pruitt (2015) constructs latent factors that explicitly target variation in the predictor set $\mathbf{X}$ which is most relevant for forecasting the target variable Y. Let L denote the number of lags used for the dependent variable. In the first pass, each predictor $X_{i,t}$ is regressed on L lags of Y to estimate its responsiveness:

$$X_{i,t} = \phi_{i,0} + \sum_{\ell=1}^{L} \phi_{i,\ell} Y_{t-\ell} + u_{i,t}, \quad i = 1, \ldots, N.$$

The estimated coefficients are collected in the matrix $\hat{\boldsymbol{\Phi}} \in \mathbb{R}^{N \times L}$, where each row represents the sensitivity of a predictor to the historical movements of Y. In the second pass, latent factors $\mathbf{F}_t$ are extracted by projecting the cross-sectional vector of predictors $\mathbf{X}_t$ onto this sensitivity matrix:

$$\mathbf{X}_t = \hat{\boldsymbol{\Phi}} \mathbf{F}_t + \boldsymbol{\epsilon}_t, \quad \hat{\mathbf{F}}_t = (\hat{\boldsymbol{\Phi}}' \hat{\boldsymbol{\Phi}})^{-1} \hat{\boldsymbol{\Phi}}' \mathbf{X}_t.$$

These factors emphasize predictors that are most responsive to Ys dynamics. In an optional third pass, additional principal components can be appended to $\mathbf{F}_t$ to capture any remaining variation orthogonal to the targeted factors. Finally, the target variable is regressed on the augmented factors:

$$\hat{Y}_{t+1|t} = \hat{\alpha}_0 + \sum_{j=1}^{k} \hat{\alpha}_j F_{j,t}^{\text{aug}}.$$

where $F_{j,t}^{\text{aug}}$ denotes the augmented factor set, and k is the total number of factors. The model is re-estimated within a rolling window to accommodate time variation in predictive relationships.

3.5.2 Principal Component LASSO/LAR (PCLAR/PCLAS)

The Principal Component LASSO/LAR approaches, introduced by Bai and Ng (2008), combine dimension reduction with variable pre-selection to extract factors that are particularly relevant for forecasting. In PCLAR, principal component analysis is applied to the full predictor matrix $\mathbf{X}$, producing factors $F_{1,t}, \ldots, F_{r,t}$ that summarize the

dominant variation in the predictors. The forecasting regression estimated within each rolling window takes the form:

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \sum_{j=1}^{r} \beta_{j+1} F_{j,t} + \varepsilon_t.$$

For generating a one-step-ahead forecast, the next-period factors are obtained by projecting the next-period predictor vector $\mathbf{X}_{t+1}$ onto the PCA loadings estimated from the window up to time t. Denoting these as $\hat{F}_{j,t+1|t}$, the forecast is constructed as:

$$\hat{Y}_{t+1|t} = \hat{\beta}_0 + \hat{\beta}_1 Y_t + \sum_{j=1}^{r} \hat{\beta}_{j+1} \hat{F}_{j,t+1|t}.$$

This approach uses the contemporaneously available predictor information at time $t+1$ while keeping the factor loadings fixed at their estimated in-sample values.

PCLAS extends this framework by first selecting an active set of predictors based on their correlation with the target variable Y_t. Only the top K predictors are retained, and PCA is applied to this reduced matrix to extract a smaller set of factors $F^{\text{AS}}_{1,t}, \ldots, F^{\text{AS}}_{r^{\text{AS}},t}$. Within each rolling window, the forecasting regression is estimated as:

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \sum_{j=1}^{r^{\text{AS}}} \beta_{j+1} F^{\text{AS}}_{j,t} + \varepsilon_t.$$

For the one-step-ahead forecast, the next-period active-set factors are computed by projecting the next-period predictor vector $\mathbf{X}_{t+1}$ (restricted to the selected active predictors) onto the PCA loadings estimated from the window up to time t. Denoting these as $\hat{F}^{\text{AS}}_{j,t+1|t}$, the forecast is given by:

$$\hat{Y}_{t+1|t} = \hat{\beta}_0 + \hat{\beta}_1 Y_t + \sum_{j=1}^{r^{\text{AS}}} \hat{\beta}_{j+1} \hat{F}^{\text{AS}}_{j,t+1|t}.$$

Both PCLAR and PCLAS determine the number of factors using a data-driven criterion such as the Bai and Ng IC_2, ensuring an optimal trade-off between goodness of fit and simplicity. By targeting predictors most informative for Y_t through either lag-based sensitivity (3PRF) or correlation-driven variable selection (PCLAS), these supervised approaches extract factors that are directly aligned with the forecasting objective, in contrast to unsupervised PCA which maximizes only variance in the predictor matrix $\mathbf{X}$.

3.6 Regularized Regression

The next class of forecasting methods used in this study are the penalized regression models. Penalized regression methods (often called shrinkage estimators) provide a practical way to deal with the large number of macroeconomic and financial predictors used in the analysis. When the dimension of the regressor set is high, standard OLS estimates can become unstable because many variables carry similar information or exhibit strong correlations. Regularization techniques address this problem by shrinking the size of the coefficients, which reduces estimation noise and helps improve out-of-sample performance. The study considers three commonly used penalties: Ridge, LASSO, and the Elastic Net.

3.6.1 Ridge

Ridge Regression introduces an L_2 penalty that shrinks the regression coefficients toward zero while retaining all predictors in the model. By minimizing the following penalized least squares objective, we obtain the estimator.

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{t=1}^{T} (Y_t - \beta_0 - \mathbf{X}_t \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{N} \beta_j^2 \right\}. \tag{7}$$

The L_2 penalty shrinks coefficients toward zero without eliminating predictors. This helps stabilize the estimates when predictors are strongly correlated and ensures that the regression remains well behaved even when N is large relative to T.

3.6.2 Least Absolute Shrinkage and Selection Operator (LASSO)

LASSO introduces an L_1 penalty on the regression coefficients, which modifies the least squares objective to

$$\hat{\boldsymbol{\beta}}^{\text{LASSO}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{t=1}^{T} (Y_t - \beta_0 - \mathbf{X}_t \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{N} |\beta_j| \right\}. \tag{8}$$

The L_1 penalty is non-differentiable at zero, and as a result the estimator forces many coefficients to be exactly zero when λ is sufficiently large. LASSO therefore performs variable selection and estimation simultaneously, producing sparse forecasting models in which only a subset of predictors is retained.

3.6.3 Elastic Net (EN)

The Elastic Net combines the L_1 and L_2 penalties in a single objective function,

$$\hat{\boldsymbol{\beta}}^{\text{EN}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{t=1}^{T} (Y_t - \beta_0 - \mathbf{X}_t \boldsymbol{\beta})^2 + \lambda \left[(1-\alpha) \sum_{j=1}^{N} \beta_j^2 + \alpha \sum_{j=1}^{N} |\beta_j| \right] \right\}. \tag{9}$$

By combining the shrinkage property of Ridge with the sparsity-inducing effect of LASSO, the Elastic Net is particularly effective when predictors are highly correlated. It can retain groups of related variables while still eliminating irrelevant ones, which is a valuable feature in macroeconomic applications where many indicators move together over time.

4 Empirical Results

4.1 Baseline Model: Random Walk Model

The Random Walk (RW) model serves as the baseline for out-of-sample forecasting, providing a strict reference point that all other models must surpass to demonstrate predictive value.

From Table 2, the RW model clearly performs the worst among all methods. Its Mean Squared Error (MSE) reaches 42.3450, and its Root Mean Squared Error (RMSE) is 6.5073–the highest in the entire comparison. More importantly, the out-of-sample R^2 is -0.2461, meaning the RW model performs worse than simply predicting the historical average. Its relative RMSE of 1.0000 emphasizes its role as a benchmark that effective model should reduce RMSE below RW's 6.5073.

Table 2: Out-of-Sample Forecasting Performance (Fixed 100-Period Rolling Window)

Model	**MSE**	**RMSE**	**MAE**	**R^2**	**Relative RMSE**
Elastic Net	15.9884	3.9986	2.9635	0.5295	0.6145
Lasso	16.4438	4.0551	2.9302	0.5161	0.6232
PCLAS	16.7198	4.0890	3.0105	0.5104	0.6284
PCR	17.3367	4.1637	3.1791	0.4898	0.6399
DFM	17.6916	4.2061	3.1994	0.4794	0.6464
PCLAR	17.7511	4.2132	3.1986	0.4803	0.6475
Ridge	18.8720	4.3442	3.2462	0.4447	0.6676
3PRF	38.0830	6.1711	4.2043	-0.0902	0.9483
Random Walk	42.3450	6.5073	5.2410	-0.2461	1.0000

In Figure 2, the forecast values consistently lag behind the actual values. This is a direct consequence of the model's single-cycle lag structure. This characteristic prevents the RW model from anticipating sudden changes or directional shifts, particularly during periods of extreme volatility like the 2008 financial crisis, where the forecast line completely fails to capture the sharp downward trend.

We recognize that despite its simplicity and widespread application, the RW model fails to capture meaningful patterns in the data. It can only replicate historical trends without extracting valuable signals. This insight suggests that in highly volatile and nonlinear environments, models must integrate richer information and incorporate explicit

forecasting mechanisms, such as factor extraction, variable selection, or regularization. Otherwise it cannot achieve good predictive performance beyond simple benchmarks.

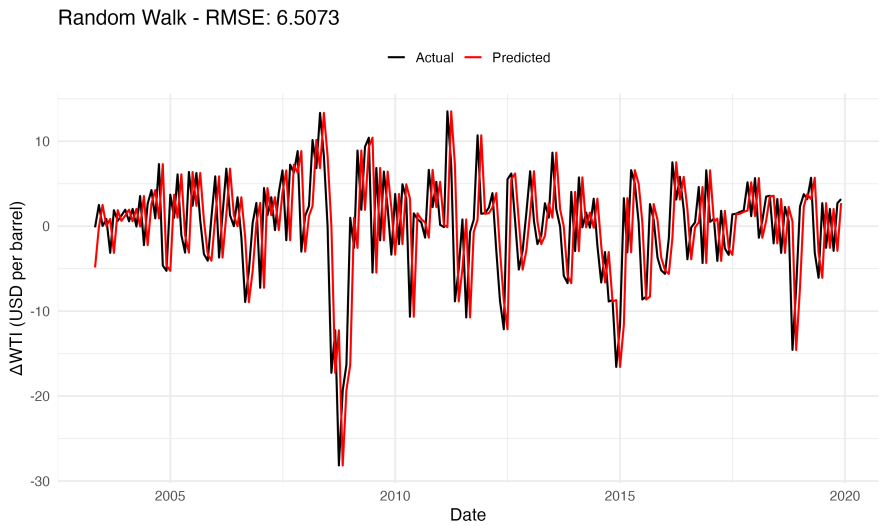

Figure 2: Random Walk

4.2 Unsupervised Factor Models

Principal Component Regression (PCR) and Dynamic Factor Model (DFM) both compress high-dimensional macroeconomic data into a few common factors. These models adopt the IC_2 criterion proposed by Bai and Ng (2002) to dynamically select the number of factors k and are re-estimated within a fixed 100-period rolling window.

4.2.1 Principal Component Regression (PCR)

From Table 2, PCR demonstrates solid performance in out-of-sample forecasting. Its RMSE is 4.1637, with a Relative RMSE of 0.6399, indicating that PCR reduces forecast error by roughly 36.01% ($1 - 0.6399 = 0.3601$) compared to the Random Walk benchmark. The out-of-sample R^2 reaches 0.4898, which suggests that a substantial part of the variation in oil prices can be explained by common macroeconomic factors extracted from the predictor set.

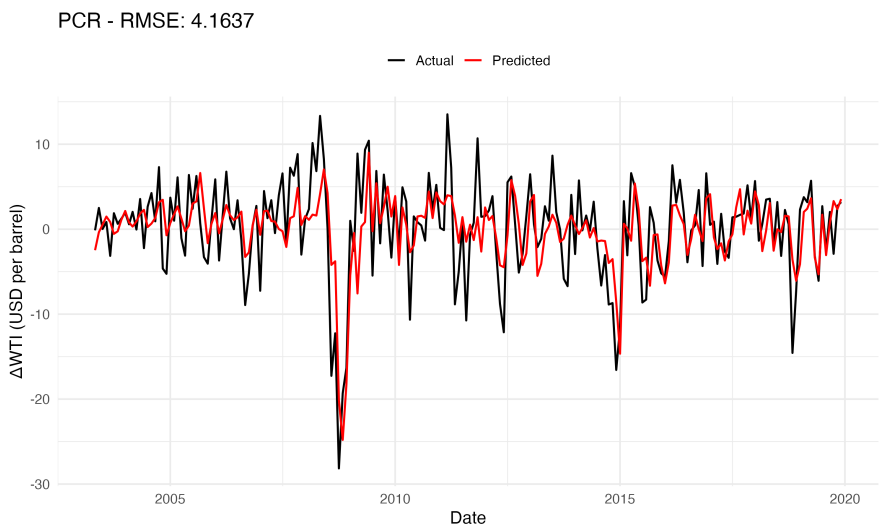

Figure 3: PCR Model Forecast vs Actual (RMSE = 4.1637)

Figure 3 shows that the PCR forecasts (red) closely follow the actual values (black), including during periods of sharp volatility such as the 2008 financial crisis. This behavior reflects the strengths of PCR in a "large N, small T" environment. In the first stage, PCA extracts a small number of factors that summarize the common movements in the predictors and substantially reduce multicollinearity. In the second stage, regressing oil price changes on these

factors avoids estimating a large set of coefficients, thereby limiting variance and stabilizing the model. These features allow PCR to capture broad macroeconomic signals that are relevant for oil price dynamics.

4.2.2 Dynamic Factor Model (DFM)

The Dynamic Factor Model (DFM) extends PCR by adding an autoregressive term, so the forecasting equation includes both the macroeconomic factors F_t and the lagged oil price Y_{t-1}. The idea is that the factors capture broad economic conditions, while oil prices often show persistence that a lagged term can help account for. The purpose of DFM is therefore to examine whether adding the oil prices own past values improves forecast performance.

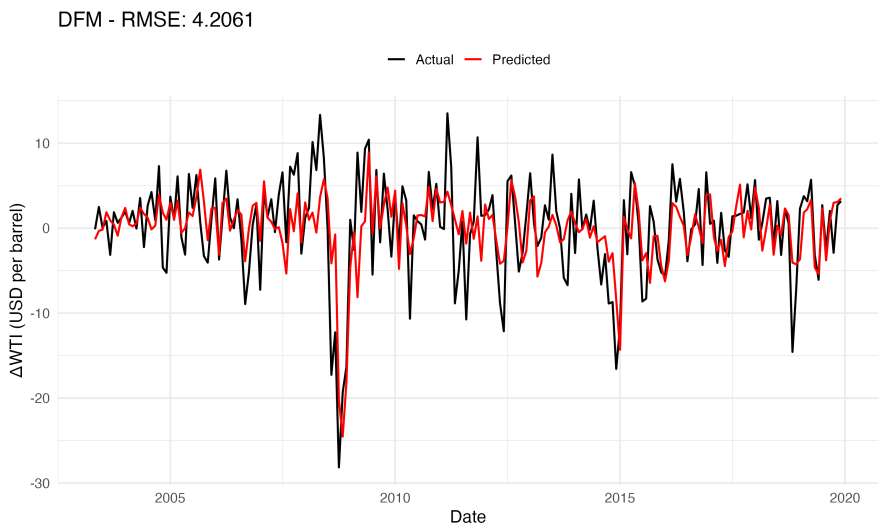

Figure 4: DFM Model Forecast vs Actual (RMSE = 4.2061)

From Table 2, DFM achieves an RMSE of 4.2061 and a Relative RMSE of 0.6464. Although this is still significantly better than the benchmark, it is slightly worse than PCR. Its R^2 (0.4794) is also lower than PCR.

Looking at Figure 4, the predicted series follows the actual series reasonably well, but not better than PCR. This raised a question for us: does adding the lagged oil price really help? Based on these results, it seems that the common factors already capture most of the predictable structure, and the autoregressive term does not provide substantial incremental gainperhaps even introducing noise. Therefore, under the current dataset and rolling window configuration, PCR appears more efficient and simpler than DFM. This finding made us realize that model complexity does not always guarantee better forecasting performance.

4.3 Supervised Factor Model

While PCR and DFM show the value of unsupervised factor models, they ignore the link between extracted factors and the prediction target. We turn to supervised approaches that incorporate the target variable into factor extraction or variable selection. The IC_2 criterion proposed by Bai and Ng (2002) to select number of factors is adopted within each rolling window.

4.3.1 Three-Pass Regression Filter (3PRF)

The results indicate that the Three-Pass Regression Filter (3PRF) exhibits limited forecasting ability in this high-dimensional setting. As shown in Table 2, its MSE (38.0830) and RMSE (6.1711) are only slightly better than the Random Walk benchmark, suggesting the presence of some weak predictive signal. However, the out-of-sample R^2 of -0.0902 demonstrates that the model performs worse than a naive forecast based on the historical mean, capturing almost none of the variation in monthly WTI price changes. The Relative RMSE of 0.9483 further emphasizes that efficiency gains are minimal compared to other methods.

Figure 5 provides a clearer illustration of the model's performance. The predicted values generally follow the overall trend of the actual data, but the model's predictive power significantly declines in the face of sudden market volatility, particularly during the 2008-2009 financial crisis. This may reflect the 3PRF model's sensitivity to lagged variable

selection and its difficulty in extracting effective target factors when the set of predictive variables contains noise or instability. Overall, despite its solid theoretical foundation, the 3PRF model's practical predictive performance is constrained.

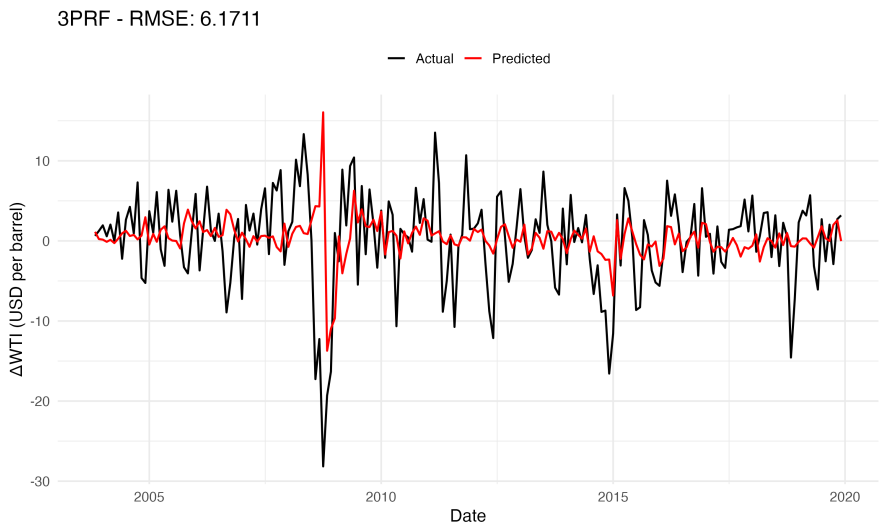

Figure 5: Three-Pass Regression Filter (3PRF) Forecast vs. Actual

4.3.2 Principal Component LASSO (PCLAR)

In contrast, the Relative RMSE of 0.6475 indicates that the Principal Component LASSO (PCLAR) model significantly outperforms the baseline model. Also, its RMSE is 4.2132, with an out-of-sample R^2 value of 0.4803. This shows that PCLAR captures more predictive signals in high-dimensional data, exhibiting significantly stronger predictive capability than 3PRF.

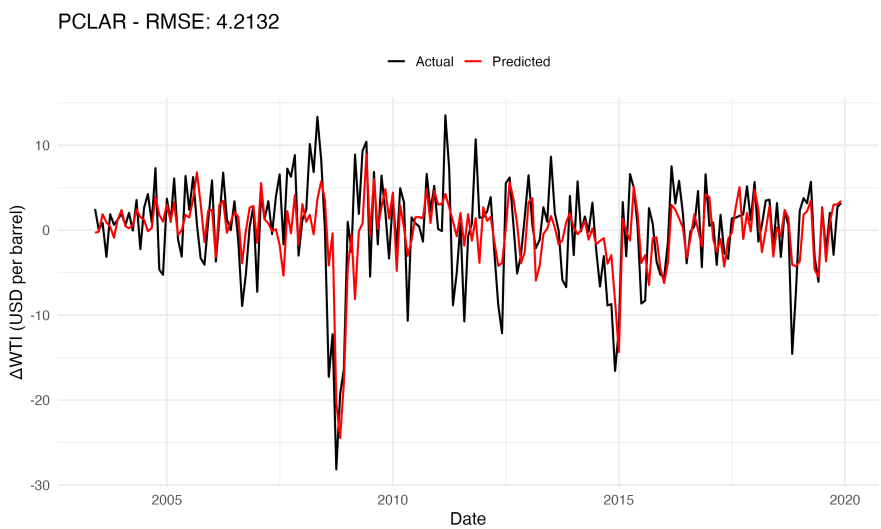

Figure 6: Principal Component LASSO (PCLAR) Forecast vs. Actual

Figure 6 displays high alignment between predicted and actual values. The model effectively captures overall trends and medium-term fluctuations while smoothing some short-term volatility. The introduction of lagged dependent variables endows predictions with persistence, significantly enhancing short-term forecasting accuracy during market turbulence. This proves that combining factor extraction with directional regularization in PCLAR improves predictive performance in high-dimensional forecasting scenarios.

4.3.3 Principal Component LASSO with Active Set (PCLAS)

The Principal Component LASSO with Active Set (PCLAS) demonstrated the most outstanding performance among supervised factor models. As displayed in Table 2, the RMSE was 4.0890, and the out-of-sample R^2 value reached 0.5104. The relative RMSE of 0.6284 indicates that PCLAS not only outperforms the random walk baseline model but also slightly surpasses the PCLAR model, showing its exceptional capability in extracting predictive signals from high-dimensional data.

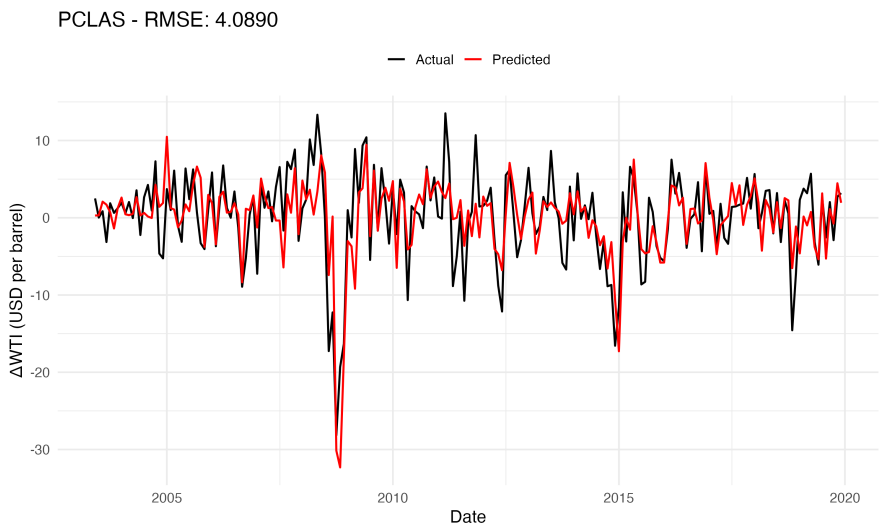

Figure 7: Principal Component LASSO with Active Set (PCLAS) Forecast vs. Actual

Figure 7 shows that the predicted values from PCLAS align closely with the actual monthly changes in WTI prices. The model accurately captures both the overall trend and medium-term fluctuations, even during periods of sudden market shifts. This improved performance is largely due to its active set selection, which removes irrelevant or noisy predictors before performing PCA. By concentrating on the most informative variables, PCLAS constructs more stable and interpretable factors, resulting in sharper, more reliable, and robust forecasts compared to other supervised factor approaches.

4.4 Regularized Regression

An alternative way to handle large sets of predictors is offered by using regularization methods. To be specific, they add penalty terms that shrink regression coefficients and reduce the risk of overfitting. Unlike factor-based approaches, these methods model the relationship between the predictors and the dependent variable directly, while imposing either an L_2, L_1, or combined penalty to control the complexity of the specification. For each rolling window, the tuning parameter (λ is chosen using the Bayesian Information Criterion (BIC), which allows the model to maintain an appropriate balance between flexibility and simplicity.

4.4.1 Ridge Regression

Ridge regression serves as a natural starting point among the regularization methods. By imposing an L_2 penalty, it shrinks the coefficients toward zero while retaining all predictors in the specification. Although this approach helps stabilize the estimates in the presence of multicollinearity, we expect that it may still preserve too much noise when the predictor set is large. As reported in Table 2, Ridge yields an MSE of 18.8720 and an RMSE of 4.3442, corresponding to a Relative RMSE of 0.6676—approximately a one-third reduction in forecast error relative to the Random Walk. Its out-of-sample R^2 of 0.4447 indicates that the model explains a meaningful share of the variation in monthly WTI price changes, though it does not match the performance achieved by the sparser regularization techniques.

A clearer picture of the models behavior is shown in Figure 8. The forecasts follow the general direction of the data but react more slowly when prices change sharply, most notably during the 2008–2009 downturn. This pattern supports our initial conjecture. Such smoothing is characteristic of the L_2 penalty: while it improves estimation stability when predictors are highly correlated, it also makes the model less responsive during periods of abrupt market movements.

4.4.2 LASSO Regression

LASSO applies an L_1 penalty, which not only shrinks the coefficients but also forces many of them to zero. Since Ridge showed limited ability to reduce dimensionality, we expect that the sparsity enforced by the L_1 penalty will yield stronger out-of-sample performance by isolating only the most informative predictors. This feature enables the model to identify a smaller, more relevant subset of variables automatically. As shown in Table 2, LASSO improves upon Ridge by producing an MSE of 16.4438 and an RMSE of 4.0551. The Relative RMSE decreases to 0.6232, and the out-of-sample R^2 increases to 0.5161, indicating that the predictors selected through the penalty carry a clearer and more meaningful predictive signal.

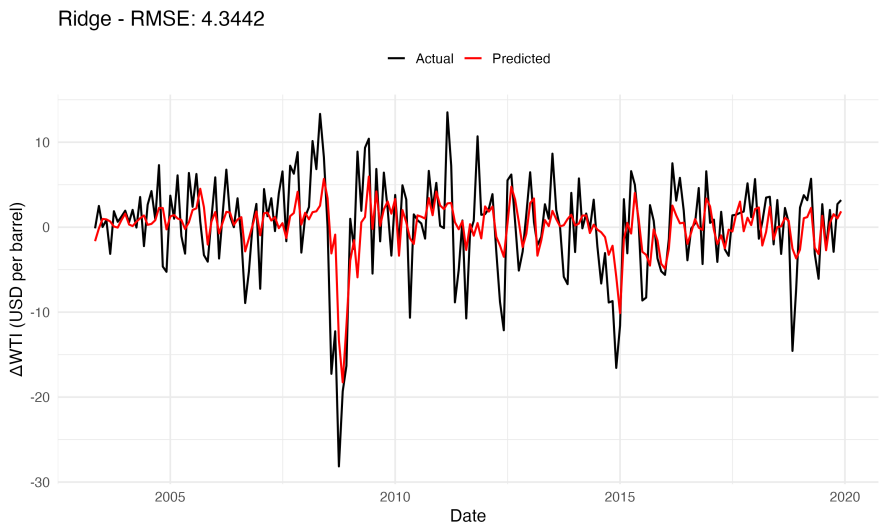

Figure 8: Ridge Forecast vs Actual (RMSE = 4.3442)

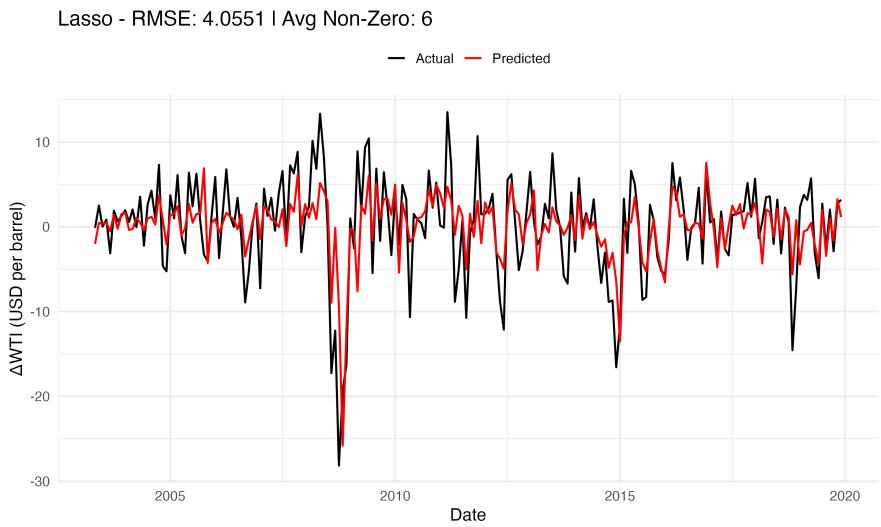

Figure 9: LASSO Forecast vs Actual (RMSE = 4.0551)

As shown in Figure 9, sparsity plays a clear role in shaping the forecast path. LASSO responds noticeably faster during periods of turbulence, and it captures several turning points that Ridge tends to delay. This is largely due to its reliance on only a handful of relevant predictors. The resulting forecasts remain smooth enough for interpretation, yet they shift direction more promptly when market conditions change.

4.4.3 Elastic Net

The Elastic Net combines the advantages of both L_1 and L_2 penalties. It can select variables while still keeping groups of correlated predictors, which is especially important in macroeconomic datasets. LASSO shows the benefit of sparsity, yet many macro variables move together, so relying only on L_1 can be unstable. For this reason, we examine

whether the additional L_2 term in the Elastic Net can steady the selection process and improve performance. As reported in Table 2, it achieves the strongest results among the regularization methods, with an MSE of 15.9884 and an RMSE of 3.9986. The Relative RMSE drops to 0.6145, a 38.55% improvement over the Random Walk, and the out-of-sample R^2 of 0.5295 is the highest in this group.

Figure 10 supports these findings. The model tracks price changes closely in both calm and turbulent periods. It reacts faster than Ridge but avoids the occasional noise amplification that appears in LASSO. This combination of stability and quick adjustment makes the Elastic Net a particularly effective tool for forecasting monthly changes in WTI prices.

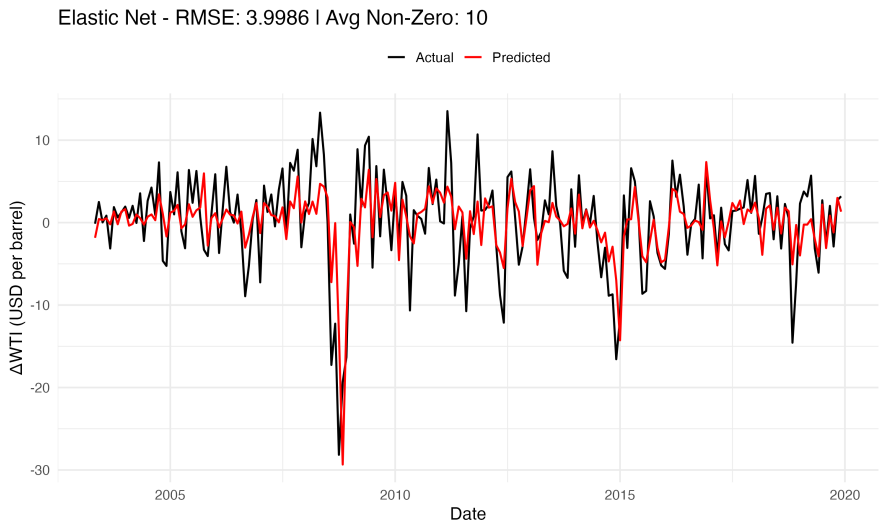

Figure 10: Elastic Net Forecast vs Actual (RMSE = 3.9986)

5 Discussion and Conclusion

The ranking of model performance shows that Elastic Net and LASSO deliver the strongest forecast accuracy, with Elastic Net achieving the lowest RMSE. Factor-based models (PCR, DFM, PCLAS, and PCLAR) perform slightly worse but still improve noticeably over the baseline Random Walk.

This comparison shows a clear progression in forecasting performance. All high-dimensional models perform consistently well, indicating that a broad macrofinancialenergy dataset captures the main drivers of oil price fluctuations. Performance improves further when variable selection is applied: L1-based methods such as LASSO and Elastic Net outperform Ridge regression, suggesting that identifying a small set of informative predictors is more important than merely shrinking coefficients in complex energy markets. Among these, Elastic Net performs best, as it combines L1-based selection with the stability of L2 regularization, allowing both effective predictor screening and robust estimation when variables are correlated.

Our analysis also reveals a key limitation. The three-pass regression filter (3PRF), though theoretically appealing and consistent under fixed-parameter assumptions in large panels (Kelly and Pruitt, 2015), performs poorly here, ranking only above the Random Walk [6]. Given the volatility of energy markets and our use of first-differenced data with short rolling windows, these relationships are likely unstable. This suggests that both factor-based and regularized high-dimensional methods must handle temporal variations, pointing to a direction for future research.

In conclusion, we compared various high-dimensional methods for forecasting monthly changes in WTI crude oil prices against a Random Walk benchmark. Regularization-based models, particularly Elastic Net, perform best, consistently reducing forecast errors. These results indicate that high-dimensional macro-financial information has predictive value when the data are stabilized and the method includes effective variable selection. Future work may explore frameworks that combine sparsity with approaches that better accommodate time-varying relationships, which are central to improving real-time oil price predictions.

[6] The limited performance of 3PRF is consistent with previous studies; for example, Guérin et al. (2020) show that 3PRF can lose efficiency when predictive relationships, like factor loadings, change over time. This lack of robustness is especially relevant in volatile energy markets.

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